



Haskell for Computational Linguists

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What are we doing today?

Introduction

Setup &
installation

Hello.hs

Phonology.hs

Syntax.hs

I will (gently) introduce you to a new paradigm, **functional programming**, and a new language, **Haskell**.

Agenda:

- 1 (10 min.) Introduction & installation
- 2 (15 min.) Recursion, thinking functionally, and Haskell syntax
- 3 (15 min.) Building a Finnish vowel harmonizer
- 4 (15 min.) Building an English → German VP translator



Who am I to tell you what to do?

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Hello y'all :) My name is Pranav. I'm a fourth-semester M.Sc. student at Universität Stuttgart, studying Computational Linguistics (specializing in Digital Phonetics).

Previously:

- **2021 - 2023:** Product Manager @ New Relic
- **2017 - 2021:** B.A. in Linguistics & Computer Science @ University of California, Los Angeles
- **1999 - 2005:** My parents' favorite child

Coding:

- **before 2017:** Python, HTML
- **since 2017:** C/C++, Java, JavaScript, Go, Haskell, OCaml, Rust



Why care about Haskell?

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- **Readable:** *Function composition* keeps code short and process-driven
- **Strong pattern-matching:** Easy to unpack and inspect custom data types
- **Elegant:** (My opinion, but) code looks clean and there's cute "syntactic sugar" everywhere

Specifically for linguists:

- **Recursion instead of iteration:** Easier to work with trees!
- **Actually used in our field:** For example, in my LING 185A: Computational Linguistics course at UCLA
 - Search "ling 185a ucla github" to find examples of previous student work, all in Haskell



What do you get out of it?

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- A new language to add to your resume!
- Finally something that isn't Python
- Friendlier build tools like `stack` and `cabal`
- Very natural syntax compared to C++, Java, and other declarative languages



What do you need today?

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- 1 Somewhere to write (VSCode, Zed...)
- 2 Somewhere to run (iTerm2, Terminal, etc...)
- 3 A free directory

If the installation does not work, you can use an online compiler, such as:

`play.haskell.org`



If you need to use an online compiler

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While I will be demonstrating code running like this...

```
→ tacos $ ghci
GHCi, version 9.4.8: https://www.haskell.org/ghc/  :? for help
ghci> :l Hello
[1 of 1] Compiling Hello                ( Hello.hs, interpreted )
Ok, one module loaded.
ghci> square 5
25
ghci> factorialList [1, 2, 3, 4, 5]
[1,2,6,24,120]
ghci> 
```



If you need to use an online compiler

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You will need to run things like this:



```
1 square :: Int -> Int
2 square x = x * x
3
4 factorial :: Int -> Int
5 factorial x
6   | x < 0    = error "(factorial) input cannot be negative"
7   | x == 0   = 1
8   | otherwise = x * (factorial (x - 1))
9
10 main :: IO ()
11 main = do
12   print (square 5)
13   print (factorial 5)
```

25
120



Installation & setup

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- 1 haskell.org/ghcup/install
- 2 “How to install” → Download GHCup

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- 1 haskell.org/ghcup/install
- 2 “How to install” → Download GHCup
- 3 Copy “...Stub.hs” files from my GitHub into your text editor
- 4 Remove the “...Stub” and edit the files directly
- 5 Test using `ghcup` in your command line



GHCi cheat sheet

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A read-evalue-print loop (REPL) for Haskell.

- 1 `$ ghci`
 - starts GHCi
- 2 `ghci> x = 5`
 - sets “x” to “5”
- 3 `ghci> x`
 - prints value of “x”
- 4 `ghci> y`
 - throws “Variable not found in scope” error
- 5 `ghci> :l Module`
 - loads file “Module.hs”
- 6 `ghci> :q`
 - quits GHCi

Let's try it!

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- 1 Create `Hello.hs`
- 2 Copy text from `HelloStub.hs` from my GitHub
- 3 Use `$ ghci` to test how you fare!

`github.com/pranavs99/tacos-workshop`



Finnish vowel harmony

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How can we model the following phenomena?

- (1) *minä asu -n tä -ssä maa -ssa*
1.SG.NOM live -1.SG PROX -INESS country -INESS

“I live this-in country-in”

“I live in this country”

- (2) *hän muista -: tuo -sta kiele -stä*
3.SG.NOM remember -3.SG DIST -ELAT language -ELAT

“they remember-s that-from language-from”

“they (sg.) remember that language”



How does Finnish vowel harmony work?

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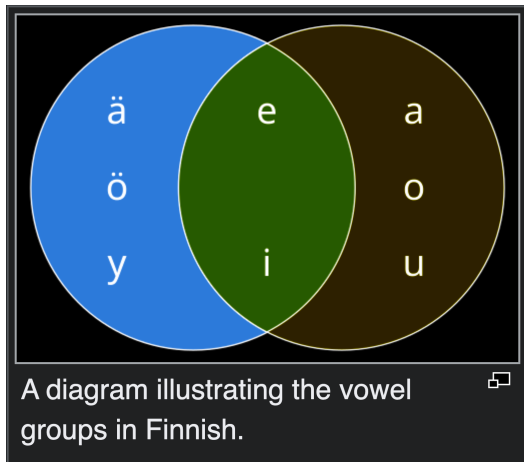
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blue = front, green = neutral, brown = back



How can we describe this as linguists?

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- Affixes comes in two forms: **front** and **back**



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- Affixes comes in two forms: **front** and **back**
- Vowels in affixes **match the frontness of the stem**



How can we describe this as linguists?

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- Affixes comes in two forms: **front** and **back**
- Vowels in affixes **match the frontness of the stem**
- When a stem only contains neutral vowels (“kiele”), the **front** affix form is used



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- Affixes comes in two forms: **front** and **back**
- Vowels in affixes **match the frontness of the stem**
- When a stem only contains neutral vowels (“kiele”), the **front** affix form is used
 - The front form is the **default**



German V2 movement

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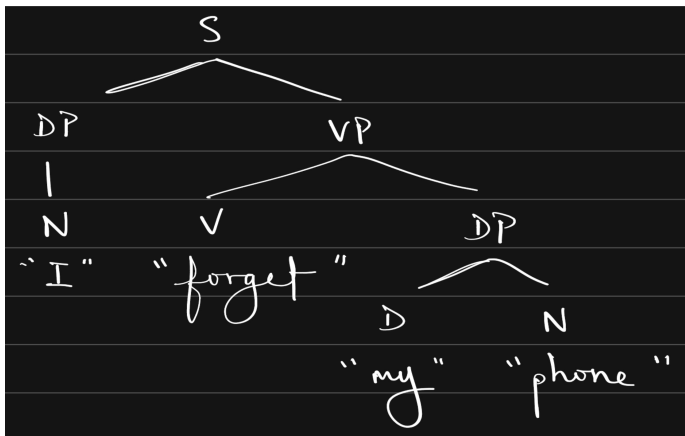
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"I forget my phone"



A hand-drawn syntax tree for the sentence "Ich vergesse mein Handy". The root node is S, which branches into DP and VP. DP branches into I, which then branches into N, with the word "ich" written below. VP branches into V, with the word "vergesse" written below, and another DP. This second DP branches into D, with the word "mein" written below, and N, with the word "Handy" written below.

```
graph TD; S --> DP1[DP]; S --> VP; DP1 --> I; I --> N1[N]; N1 --- ich["ich"]; VP --> V; V --- vergesse["vergesse"]; VP --> DP2[DP]; DP2 --> D; D --- mein["mein"]; DP2 --> N2[N]; N2 --- Handy["Handy"]
```



German V2 movement

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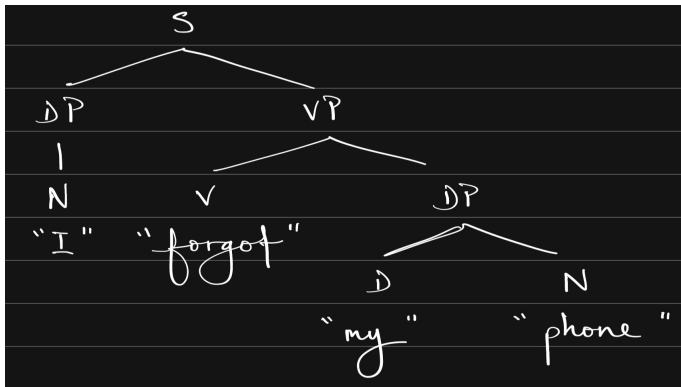
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"I forgot my phone"



German V2 movement

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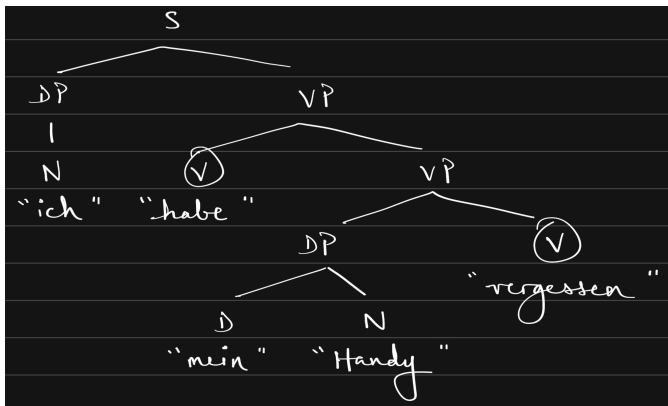
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“I **forgot** my phone” → “ich **habe** mein Handy **vergessen**”



Thank you!

